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Understanding Farmer preferences through citizen Science: Insights on potato varieties in Nigeria

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ABSTRACT

Conventional top-down potato breeding often overlooks farmer input, leading to varieties that may not be suited to local conditions, limiting their adoption and impact on smallholder farming. To address this gap, this study evaluated whether a participatory, digitally enabled approach to variety selection better reflects farmer needs and improves uptake. This study employed the tricot (Triadic Comparisons of Technology Options) approach, a participatory framework, combined with digital tools for real-time data collection and decision support. Farmers from seven Local Government Areas (LGAs) in Plateau State, Nigeria (Barikin Ladi, Bassa, Bokkos, Jos North, Jos South, Mangu, and Pankshin), evaluated anonymized potato genotypes alongside the check variety Marabel. Trial design and genotype allocation were managed through ClimMob, a cloud-based platform (https://climmob.net), enabling live streaming and real-time tracking. Data were collected via a mobile app and Open Data Kit (ODK). Assessments occurred at three crop stages: vegetative (germination, disease resistance, drought tolerance), harvest (yield, tuber appearance, marketability), and post-harvest (cooking quality, storability). Farmer preferences were analyzed using the Plackett-Luce model, a probabilistic model that estimates the relative 'worth' (also referred to as log-worth) of each genotype based on observed rankings across multiple comparisons. CIP clones CIP392797.22 and CIP393371.157 emerged as top-ranked, with log-worth values of 0.620 and 0.309, respectively, indicating stronger preference relative to the check. In contrast, CIP398208.29 and CIP398190.200 were least preferred (−0.085 and −0.465). Socioeconomic factors influenced selection: market-oriented farmers prioritized commercial traits, while subsistence farmers emphasized resilience. This study highlights the value of participatory, data-driven breeding that integrates local knowledge to enhance adoption, resilience, and food security.

1. Introduction

Potato is rapidly emerging as a key crop in Nigeria, a country historically known for its staple root and tuber crops like cassava and yam (FAOSTAT, 2023; [1]). The crop holds significant potential to contribute to both food security and income generation among smallholder farmers [1–3]. Although production is primarily concentrated in Plateau State, its expansion into other suitable agroecological zones highlights the growing national importance of the crop [4]. However, this growth is hindered by persistent challenges, including limited access to high-quality seed, insufficient availability of improved, adapted, and disease-tolerant varieties, and environmental constraints such as declining soil fertility and erratic rainfall patterns [1,5–8].

Efforts to address these production constraints are further undermined by conventional breeding and variety release systems that often fail to capture the diverse and evolving needs of Nigerian farmers. Formal release processes involve multi-year trials, including DUS (Distinctness, Uniformity, and Stability) evaluations to assess candidate varieties [9]. However, they remain largely top-down, limiting farmer participation and local knowledge integration. This reduces the

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relevance of released varieties and contributes to low adoption rates—especially in the public sector where market promotion is minimal [10]. Consequently, breeding efforts frequently fail to align with farmer priorities [11], a challenge that is particularly critical for vegetatively propagated crops like potatoes, where timely varietal turnover is essential to sustain productivity and resilience [12].

Recognizing these gaps, early applied research efforts emphasized the need to reposition farmers not as end-users but as co-creators in varietal development. The "Farmer-Back-to-Farmer" model exemplified this shift, linking diagnostic work with farmers directly to the design and refinement of solutions based on their lived realities [13]. This participatory foundation informed later innovations like participatory varietal selection (PVS) and the mother-baby trial approach, which improved farmer involvement in variety testing [14,15]. Yet, these models often limited participation to the evaluation of pre-selected lines, failing to embed farmer priorities early in the breeding pipeline [16]. More recent frameworks, such as CGIAR's Target Product Profiles (TPPs), attempt to institutionalize user-centered design by linking trait preferences to specific market segments [17,18].

Despite these advances, integrating such approaches into formal breeding remains difficult in smallholder settings like Nigeria, where limited technical capacity, socio-cultural barriers, and behavioral dimensions of adoption constrain farmer engagement, particularly for crops like potato where rapid varietal turnover is vital. Thus, while participatory strategies hold promise for aligning breeding with farmer realities, their impact depends on overcoming both structural and institutional inertia within conventional research systems. Many of these participatory approaches, while conceptually robust, continue to face limitations in scalability, data reliability, and their ability to capture the socio-economic and agroecological diversity typical of smallholder farming systems [19,20].

The tricot (triadic comparisons of technology options) approach offers an innovative alternative that directly addresses these constraints. By employing a crowdsourcing model, tricot transforms farmers into "citizen scientists" —non-professional researchers who systematically collect and share data from their own fields thereby contributing to varietal evaluation through decentralized, farmer-managed trials [21]. This model has shown promise across diverse applications, including the evaluation of cassava clones [22], consumer preferences for boiled sweet potato [23], participatory yield estimates for common bean [24,25], and cassava gari processing quality [26].

Unlike traditional participatory models that rely on centrally coordinated field trials, tricot leverages digital platforms to distribute small, randomized sets of three varieties per farmer—reducing cognitive load and enabling statistically robust comparisons. These sets are stratified across different agroecological zones and delivered as uniquely coded trial kits, allowing localized assessments under farmers' own conditions. Digital data capture through the ClimMob platform (https://climmob.net, [27]), facilitates real-time collection and processing using ranking-based statistical models [28], generating rigorous and actionable insights. This digital infrastructure allows researchers to efficiently identify farmer-preferred varieties with broader relevance, thereby supporting more inclusive and adaptive breeding decisions.

Tricot also excels in integrating socio-economic data, including gender-disaggregated analyses, and aligns with emerging paradigms in precision breeding and smart agriculture [29,30]. Its ability to combine farmer perspectives with machine learning and computational analysis has been shown to improve variety targeting, yield stability, and economic returns [31–33].

However, while tricot offers significant potential to democratize breeding and accelerate varietal turnover, its integration into formal breeding frameworks is not without challenges. Practical concerns such as limited digital infrastructure, uneven farmer digital literacy, unequal access to mobile technology, and institutional resistance to decentralized data can hinder widespread adoption. Many breeding systems remain embedded in developmental norms that favor standardized, station-based evaluations over dynamic, farmer-led insights. Therefore, although digital participatory approaches like tricot represent a pragmatic response to long-standing constraints in variety development, their broader applicability and institutional uptake remain uncertain. Addressing these limitations will require not only technological innovation but also systemic support, adaptive capacity, and shifts in governance, capacity building, and institutional mindsets to ensure such participatory approaches can effectively complement and enrich formal breeding pipelines.

In this study, we applied the tricot approach to assess six potato clones introduced by the International Potato Center (CIP) in Nigeria, four of which were subsequently approved and released by the National Variety Release Committee [5]. The study had two main objectives: first, to evaluate the major factors influencing farmers' variety selection using digital platforms and statistical modeling; and second, to determine whether the tricot approach provides unique, context-specific insights that can complement or enhance existing breeding objectives and variety release protocols. Additionally, the study aimed to derive lessons that could inform policy adjustments in Nigeria and other developing countries, reinforcing the role of digital technologies and electronic decision-support systems in modern agricultural research.

2. Methodology

2.1. Study site

Participants for this study were drawn from seven Local Government Areas (LGAs) in Plateau State, Nigeria: Barikin Ladi, Bassa, Bokkos, Jos North, Jos South, Mangu, and Pankshin (Fig. 1). Plateau State is well-suited for potato farming due to its cool climate, and favorable soils. The region lies at an elevation of 1200–1800 m above sea level, providing cool temperatures ranging between 18 °C and 25 °C [34]. These conditions are optimal for potato growth, minimizing heat stress and promoting tuber formation. The area receives annual rainfall of 1200–1600 mm, predominantly from April to October, with a distinct dry season between November and March [35]. This rainfall pattern supports rain-fed potato cultivation during the wet season, while the dry season provides favorable conditions for storage.

Soils in Plateau State vary across LGAs but generally support potato production [34]. Barikin Ladi and Bokkos are characterized by deep, well-drained loamy soils with high organic matter, essential for tuber development. In contrast, Jos North and Jos South feature sandy-loam soils with moderate fertility, often supplemented by fertilization to enhance productivity. Mangu and Pankshin, located in transitional zones with clay-loam and sandy soils, provide additional variability in the trial environment.

2.2. Trial design

The study employed the triadic comparison of technologies (tricot) approach [36], which incorporates a balanced incomplete block design (BIBD) to evaluate farmer preferences for technological packages based on the worst-best ranking criterion. Fig. 2 summarizes the traits ranked in this manner; a detailed list is provided in Supplementary Material Table S3 A-C. The trial included six potato genotypes—CIP392797.22, CIP393371.157, CIP393371.58, CIP381381.13, CIP398208.29, and CIP398190.200, along with the popular variety Marabel. Details of these entries and their distribution across Local Government Areas (LGA) are presented in Table 1. These genotypes were tested with 300 farmers from seven LGAs in Plateau State, Nigeria. Five farmers were excluded from the analysis due to incomplete or inconsistent ranking responses across the four crop stages or missing mandatory socioeconomic data. The ClimMob platform's quality control checks flagged these records for skipped ranking tasks or unsubmitted required fields, preventing completion of the survey. Consequently, the final dataset included 295 farmers with validated data.

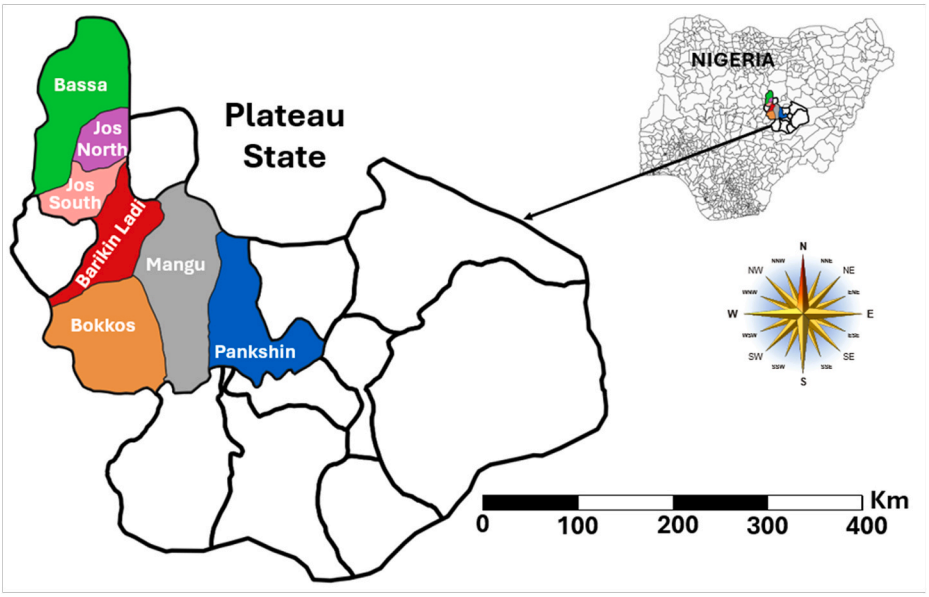


Fig. 1. Map of sampled local government areas (LGAs) in plateau State, Nigeria, for the tricot citizen science study.

The farmers served as the incomplete blocks, each receiving three out of the seven genotypes. The three genotypes per block were assigned blindly as option packages; Option A, Option B, and Option C, with each farmer evaluating only the three genotypes included in their respective package. The distribution of genotypes was balanced across farmers, regions, and demographic characteristics. This balanced assignment helped to control for potential confounding effects due to farmer background or location. The trial design and genotype allocation to each incomplete block was performed in the ClimMob platform (https://climmob.net, [27]).

2.3. Trial setup and management

Potato farmers from diverse socioeconomic and demographic backgrounds were recruited through collaboration with local extension agents, agricultural organizations, and community leaders. To mitigate recruitment bias from extension-based sampling, multiple channels were employed, including community meetings, farmer groups, and snowball sampling, to ensure the inclusion of both lead and less engaged farmers for broader representativeness. Comprehensive socioeconomic profiling enabled post-hoc stratification and statistical control of sampling bias. Standardized training was provided to farmers and extension agents to minimize variability, with agents' roles limited to supervision and data collection to reduce their influence on farmer decisions.

Community meetings were held at central locations in each LGA, where farmers were briefed on trial objectives, methodologies, and their roles in the process. See Supplementary Material Table S2 A-C for participant demographics, farming characteristics, and tricot trial experience. Each participating household received a trial kit containing three anonymized sets of certified sprouted seed tubers, labeled as Options A, B, and C, with 40 tubers in each set. This anonymization, combined with random allocation of treatment options, was implemented to minimize preferential treatment and reporting bias. Additionally, each kit included 1.8 kg of NPK fertilizer, an observation card, and a brochure outlining planting and management instructions. Farmers planted the tubers in three plots with a spacing of 0.75 m between rows and 0.3 m within rows (9 m^2 each). Farmers used their routine management practices but were required to apply them uniformly across the plots to maintain trial consistency.

2.4. Data collection

Data collection followed four crop stages: vegetative stage 1, vegetative stage 2, harvest stage, and postharvest stage (Fig. 2; Supplementary Material Table S3 A-C). During vegetative stage 1, farmers assessed germination rates, pest and disease resistance, and tolerance to drought and flooding. In vegetative stage 2, farmers continued evaluations, focusing on disease resistance, pest resistance, and drought tolerance. The harvest stage data included assessments of bacterial wilt (BW) diseases damage, days to maturity, yield, tuber size, tuber appearance, and marketability of fresh produce. In the postharvest stage, evaluations focused on consumer preferences, including ease of sale, dormancy duration, and taste after cooking. Each stage was accompanied by farmers' overall ranking of the genotypes, which provided comprehensive insights into each genotype's characteristics across different growth and storage stages.

To promote comparability across sites and participants, a standardized planting protocol was provided to all farmers. Land preparation involved flatbeds measuring 4 m × 1 m, with ridging at 75 cm intervals. Seed was planted at 10 cm depth with one seed per hill. Fertilizer application followed a uniform regime of 200 kg ha^{-1} NPK (15-15-15) at planting and 50 kg ha^{-1} urea as a top-dress at 30 days after planting. Extension agents supervised implementation of these protocols and documented any deviations, which were recorded and considered in subsequent analyses.

To account for differences in environmental conditions across farms, GPS coordinates were collected for each trial site, enabling spatial analysis and the incorporation of location-based variability into the interpretation of farmer responses. Where available, agroecological data such as rainfall patterns and altitude were used to contextualize observed performance differences.

In each evaluation, farmers compared the package set: Option A, Option B, and Option C, selecting the genotype with the best and worst traits for each characteristic, with the middle-ranked genotype inferred, resulting in a complete ranking. This data, along with socioeconomic, demographic, and GPS coordinates for spatial analysis, was recorded by extension agents using structured questionnaires on the Open Data Kit (ODK) platform, with the ClimMob platform (https://climmob.net, [27]) facilitating streamlined entry and analysis.

The socioeconomic and demographic data included participant name, age, gender, education level, degree attained, years of schooling,

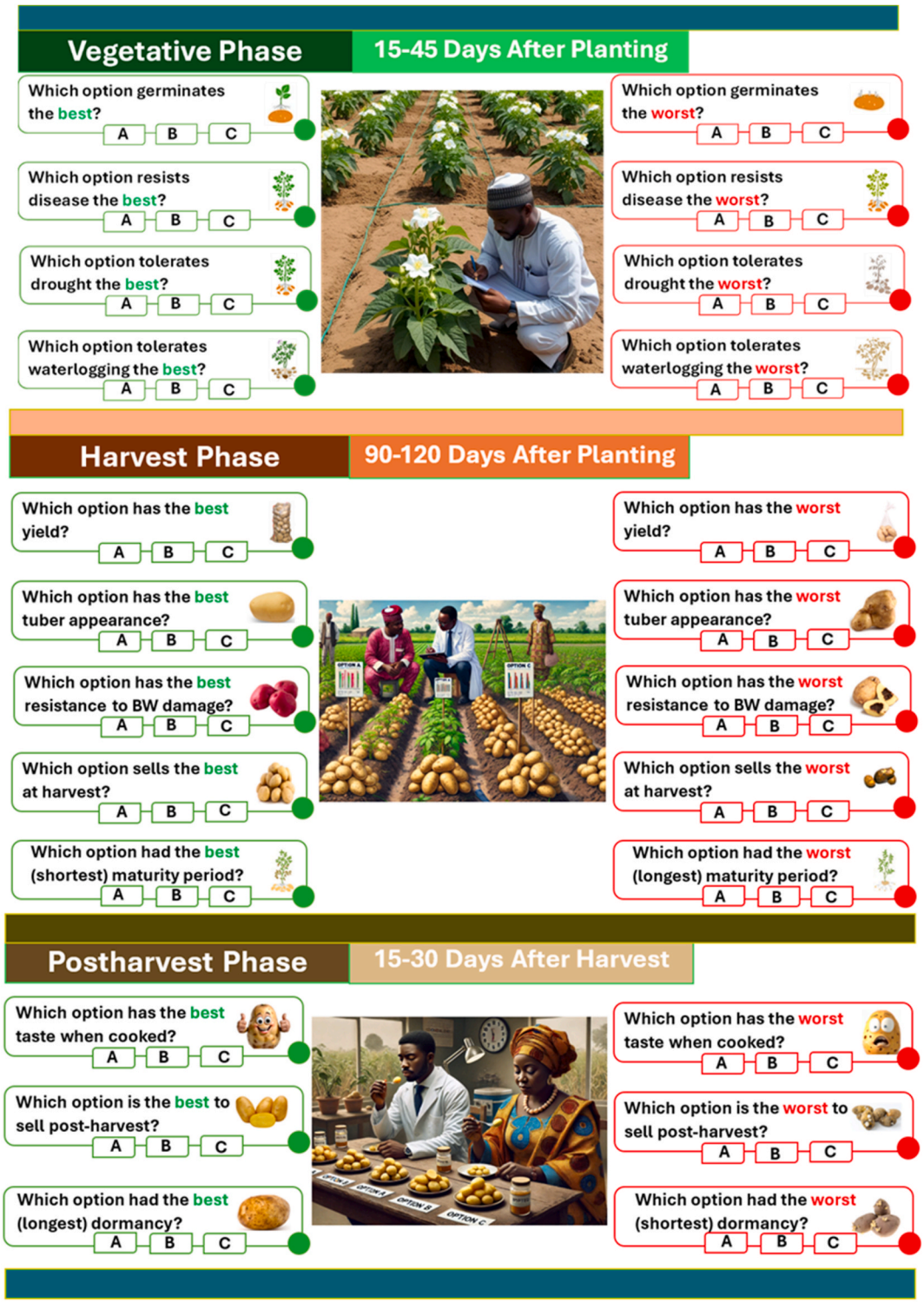


Fig. 2. Worst-best trait questions posed to farmers in tricot-evaluation stages.

marital status, occupation, household type, and size, crop selling methods, distance to markets, percentage of produce sold, contribution of produce sales to the household budget, primary decision-maker for production, organizational affiliation, contact with extension services, planting date, previous land use, land preparation methods, plot topography, and plot size. The complete datasets included in the study are presented in Supplementary Material Table S2A–C.

2.5. Data analysis

2.5.1. Statistical model

Data analyses were conducted in R [37] using the Plackett-Luce model [38], designed for ranking data where participants order items by preference. The model estimates a "worth parameter" for each item, representing the probability that the item is preferred over others. Worth parameters are expressed on a logarithmic scale (log-worth), defined as the negative logarithm of the p-value from statistical tests. Higher

Table 1
Genotype distribution across local government areas (LGAs) in plateau state, Nigeria (n = 295 farmers).

Genotype	Type	Local Name	Source	Year of Release	Barikin Ladi	Bassa	Bokkos	Jos North	Jos South	Mangu	Pankshin	Total
CIP381381.13	Test	–	CIP	–	21	9	35	8	9	28	17	127
CIP392797.22	Test	Unica	CIP	2023	22	9	34	8	9	29	17	128
CIP393371.157	Test	Juriya	CIP	2023	22	8	33	9	8	29	17	126
CIP393371.58	Test	Babban	CIP	2023	22	8	34	9	8	28	17	126
CIP398190.200	Test	Kyau	CIP	2023	21	9	33	8	9	29	18	127
CIP398208.29	Test	–	CIP	–	21	8	32	9	9	28	17	124
Marabel	Check		Europlant	2016	21	9	33	9	8	30	17	127
Total Farmers					50	20	78	20	20	67	40	295

log-worth values indicate stronger farmer preferences. The model was implemented using the PlackettLuce package [28]. In this study, higher log-worth values indicate stronger farmer preference relative to the check genotype, Marabel. To control for variability introduced by heterogenous conditions across farms, Plackett-Luce trees were used. These allowed recursive partitioning of the data into subgroups with statistically distinct ranking patterns based on covariates such as agro-ecological zone, farm management type, or household characteristics. Each terminal node was modeled with an independent Plackett-Luce model, thus allowing for context-specific preference estimation.

2.5.2. Trait selection, CART analysis, and Farmer segmentation

Trait selection was guided by Kendall's Tau correlation [39] to assess associations with genotype rankings, using the correlation function with method = 'kendall'. Traits with low contributions were iteratively removed via backward elimination using stepAIC (MASS package), retaining those with significantly high log-worth values. To identify key socio-economic factors influencing farmer preferences, a Classification and Regression Tree (CART) was constructed using the rpart package (Therneau and Atkinson [40]. Significant cutoffs were identified for household size (6.7 members), age (38.2 years), percentage of potatoes sold (42.4 %), and distance to market (8 km), as detailed in Supplementary Tables S2 and S3. Although gender and education were initially considered, they showed no significant associations with genotype rankings and were excluded from the final model, as splits based on these variables in the CART analysis did not reduce model impurity, indicating limited explanatory power in predicting farmer preferences.

Using these key socio-economic variables, farmer cohorts were defined through a two-step approach. First, the CART-significant factors—household size, age, percentage of potatoes sold, and distance to market, were used to identify the most relevant factors. This was followed by K-means clustering using the *cluster* package [41] to group farmers accordingly. The resulting clusters were descriptively labeled using a large language model (Hugging Face API), which synthesized traits and preferences into intuitive cohort descriptors. This labeling was used solely for interpretation and not for statistical inference. A likelihood-ratio test (Gosset package) was employed to assess differences in trait preferences across cohorts, while chi-square tests were used to evaluate the effect of covariates on model fit. Environmental metadata (e.g., GPS, altitude) were used post hoc to interpret spatial alignment with identified cohorts.

2.5.3. Principal Component Analysis (PCA)

PCA was performed using the prcomp function to visualize relationships between genotypes and trait preferences, based on the traits selected via Kendall filtering. Trait values were scaled, and biplots were generated using ggplot2. Genotype clusters reflected shared performance patterns, while vectors showed trait influence. Sampling adequacy for PCA was assessed using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's test of sphericity (via the psych package), confirming suitability for dimensionality reduction (KMO >0.6, $p < 0.001$). Separate PCA runs were conducted for each farmer cohort and for each production stage (vegetative, harvest, postharvest, overall). In a secondary segmentation analysis, age and household size were held constant to isolate the influence of commercial orientation, leading to a binary classification of "business-oriented" versus "subsistence-oriented" farmers. To ensure statistical robustness, the number of retained components was guided by the Kaiser criterion (eigenvalues >1) and scree plot inspection, while 10-fold cross-validation was used to assess the stability of PCA loadings. In addition, bootstrap confidence ellipses (n = 1000) were used to evaluate the reproducibility of genotype positions and trait vectors in the PCA space. To further quantify differences between subgroups, a permutation-based multivariate analysis of variance (PERMANOVA), using the adonis function from the vegan package, was applied to test for significant divergence in trait prioritization across PCA clusters ($p < 0.05$). Procrustes rotation tests were also employed to assess the alignment of trait loading structures across cohorts, confirming consistency in the dimensional reduction space despite underlying demographic heterogeneity.

2.5.4. Stage-wise trait significance

Stage-wise weighting allowed for time-specific prioritization, capturing dynamic changes in trait importance across the production cycle. Best-worst scaling was used to assess trait importance at four evaluation stages: vegetative, harvest, postharvest, and overall. Trait significance was represented by log-worth values (–log p), with positive values denoting preference. These were normalized to compute relative weights per trait per stage using Eq. (1):

$$Weight_i = \frac{logworth_i}{\sum logworth_{all\ traits}} \quad \text{Eq. (1)}$$

where logworth*i* represents the log-worth value for trait *i* in a specific stage, and the denominator is the sum of all log-worth values for that stage. Eight coded traits were tracked across stages, cohorts, and preference direction (best vs. worst). "Good germination," though dropped earlier, was reintroduced for one genotype (CIP398190.200) due to its relevance in stage-wise analysis. Traits ranked highly across stages received greater weights in the final aggregated scores. To ensure that stage-specific preferences were not driven by spurious variation, we implemented bootstrapping (1000 resamples) of rankings per stage to assess stability of log-worth values.

3. Results

3.1. Farmer preferences for genotypes

Farmers exhibited significantly distinct preferences for the evaluated genotypes (Fig. 3). Marabel was used as the reference genotype when calculating the coefficients, with a log-worth of 0.000. Relative to Marabel, CIP392797.22 showed the highest preference with a log-worth of 0.620 (SE = 0.072, $z = 8.61$, $p < 0.001$), indicating strong favorability. CIP393371.157 followed with a positive preference of 0.311 (SE = 0.074, $z = 4.20$, $p < 0.001$). Moderate preference was observed for CIP393371.58, which had a log-worth of 0.167 (SE = 0.076, $z = 2.20$, $p = 0.028$). Genotype CIP381381.13 had a marginally positive preference, with a log-worth of 0.050 (SE = 0.078, $z = 0.64$, $p = 0.522$), indicating non-significance. In contrast, CIP398208.29 was less preferred with a

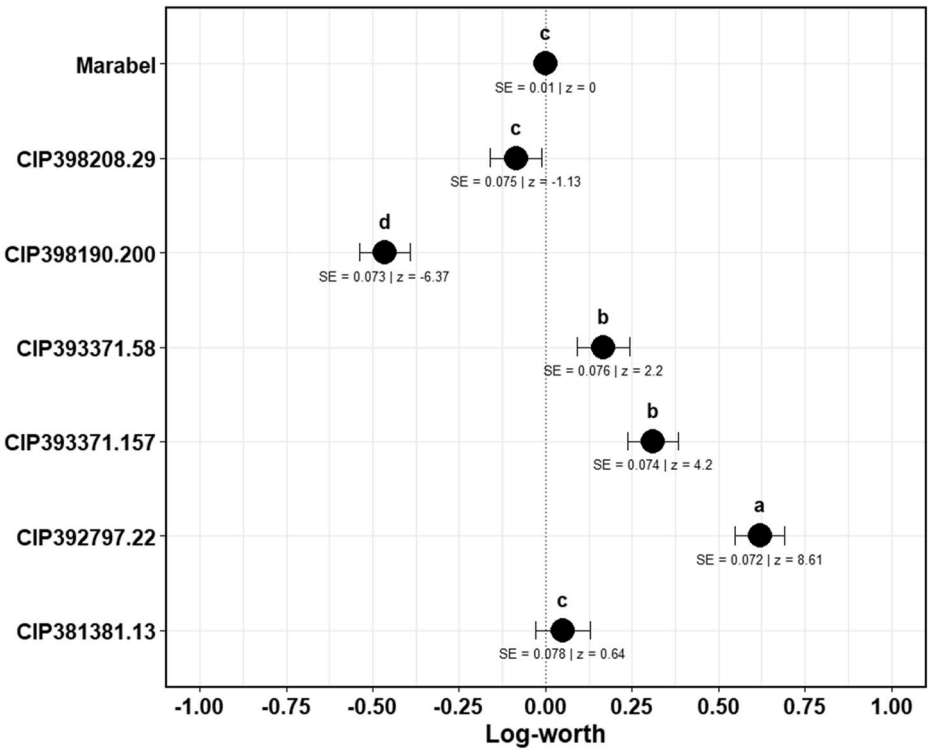


Fig. 3. Plackett-Luce estimates (log-worth) for potato genotypes tested in the study. Error bars represent quasi-standard errors, with a dashed vertical line indicating the reference point, Marabel set to zero.

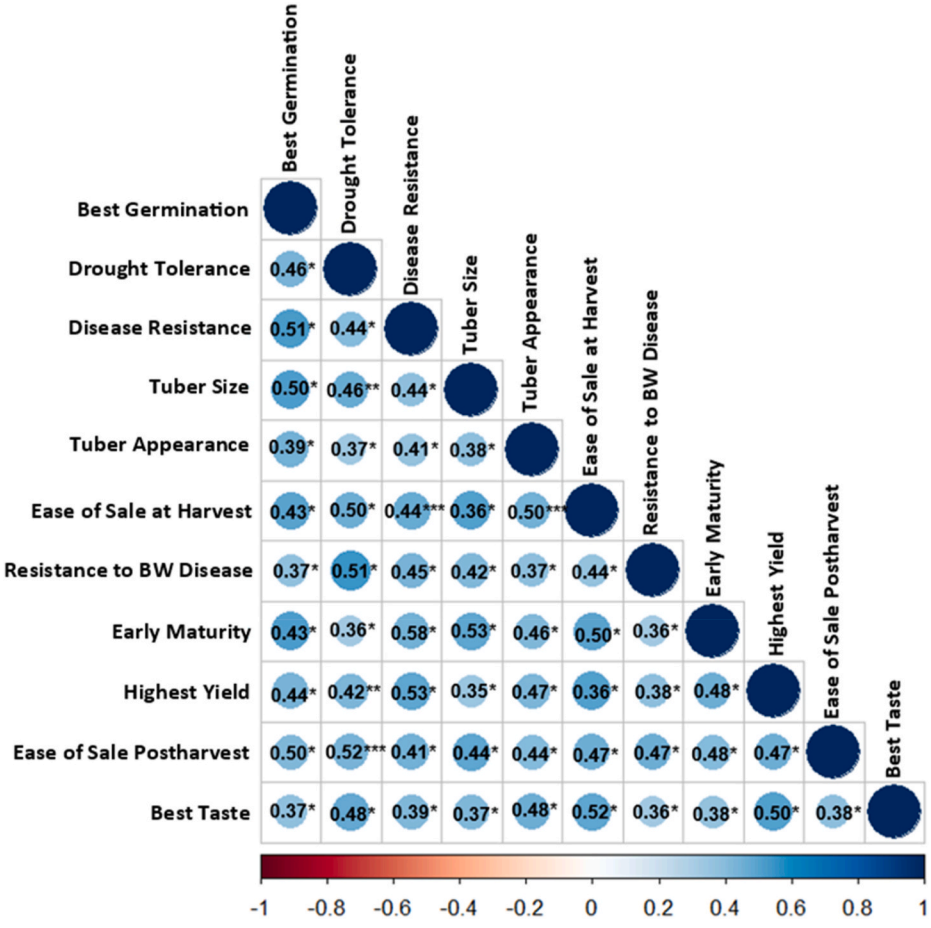


Fig. 4. Kendall correlation Matrix of potato traits from the initial iteration.

log-worth of −0.085 (SE = 0.075, $z = -1.13$, $p = 0.259$), while CIP398190.200 was significantly less preferred with a log-worth of −0.465 (SE = 0.073, $z = -6.37$, $p < 0.001$), suggesting strong aversion compared to Marabel.

3.2. Traits selection by population

A significant positive association was observed among trait factors across various crop stages. The initial feature iteration retained with moderate to strong correlations ($\tau = 0.36$ to 0.58; Fig. 4). These included three traits at vegetative stage 2 (germination, disease resistance, and drought tolerance), six traits at harvest (yield, marketability, early maturity, and resistance to bacterial wilt), and two traits in the post-harvest stage (marketability, and taste when cooked).

A subsequent, more stringent selection process addressed collinearity and model impact, removing best germination, largest tuber size, and best tuber appearance (Fig. 5). The final key traits most strongly associated with farmers' overall preference, ranked by Kendall Tau partial correlation, were: ease of sale postharvest, highest yield at harvest, disease resistance and drought tolerance at vegetative stage 2, best taste when cooked postharvest, least bacterial wilt damage at harvest, and early maturity. Traits with nonsignificant correlations ($p > 0.05$) were excluded through backward selection. Key traits were retained in over 90 % of bootstrap samples ($p < 0.05$), with Kendall Tau correlations above 0.8 confirming stable and reliable trait rankings.

3.3. Principal traits by genotype

Principal component analysis (PCA) revealed strong alignment between overall farmer preferences and the genotypes CIP392797.22 and CIP393371.157 (Fig. 6). CIP392797.22 aligned with vectors for drought tolerance at vegetative stage 2, least bacterial wilt damage at harvest, and best taste when cooked postharvest. CIP393371.157 showed strong associations with ease of sale postharvest, high yield at harvest, and disease resistance at vegetative stage 2.

The early maturity vector aligned most closely with CIP392797.22 and Marabel. In contrast, genotypes CIP381381.13, CIP398190.200, and CIP398208.29 were located in the southwestern quadrant, oriented away from dominant trait vectors, indicating minimal association with prioritized traits.

Trait–genotype associations remained consistent across farmer subgroups, with bootstrap validation (1000 iterations) confirming stability in trait loadings (retained in >90 % of samples, $p < 0.05$), supporting result robustness and generalizability.

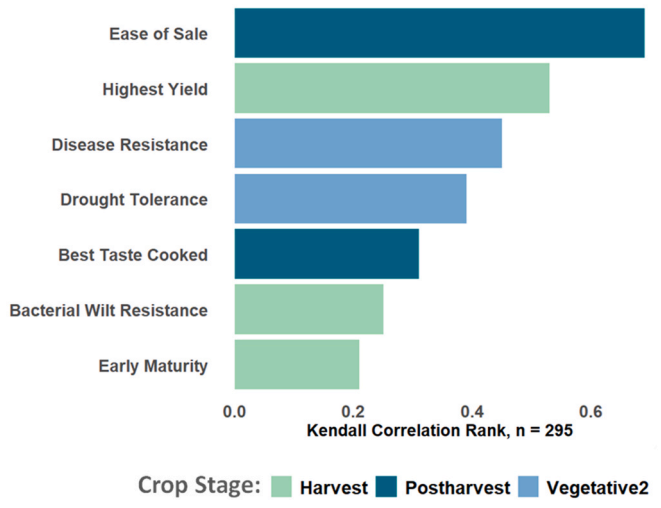


Fig. 5. Traits retained for genotype selection in the kendall second round of iterations.

3.4. Socio-economic factors in Genotype Selection for cohort segmentation

CART analysis identified 38.2 years as the age threshold, dividing farmers into two cohorts (Fig. 7).

In the <38.2 years group, genotype preference was first split by percentage of potatoes sold. Farmers selling <42.4 % retained CIP392797.22 and CIP393371.157. Within this subgroup, household size <6.1 retained CIP392797.22, while ≥6.1 retained CIP393371.157.

Among farmers selling ≥42.4 %, Marabel and CIP393371.58 were retained. Those located <8 km from market retained both genotypes; those ≥8 km retained only CIP393371.58.

In the ≥38.2 years group, household size <6.7 retained CIP381381.13 and CIP398208.29, while ≥6.7 retained CIP398190.200. Within the <6.7 subgroup, education level further differentiated preference: those with less than high school retained CIP381381.13; those with high school or higher retained CIP398208.29.

Genotype splits were confirmed in over 93 % of 1000 bootstrap samples ($p < 0.05$), indicating strong model stability and subgroup-level consistency.

3.5. Segmentation of farmers by principal socio-economic thresholds

The segmentation analysis, guided by CART regression initial cut-offs for key factors; household size (6.7), age (38.2), education level (high school), percentage of potatoes sold (42.4 %), and distance to market (8 km) resulted in the classification of farmers into four distinct cohorts based on age, business involvement, and household size (Table 2). Cohort A consisted of young farmers (≤38.2 years) from small households (≤6.7 members) actively engaged in the potato business, representing 8 % of the total sample. Cohort B included young farmers (≤38.2 years) from small households who are not involved in the potato business, accounting for 6 %. Cohort C comprised older farmers (>38.2 years) with large households (>6.7 members) who are engaged in the potato business, comprising 37 %. Cohort D consisted of older farmers from large households who did not participate in the potato business, making up 14 %. These segments were robust across bootstrap resampling iterations (n = 1000), with consistent node splits and cohort membership stability exceeding 90 %, confirming the reliability of the subgroup structure.

3.5.1. Likelihood ratio test by Cohort and overall

The likelihood-ratio test results (Table 3) revealed varying preferences across cohorts. In the Younger, Small Household in Business Cohort (Cohort A), significant traits included Ease of Sale, High Yield, and Resistance to Bacterial Wilt. For the Younger, Small Household Not in Business — Cohort B, the significant traits were; Ease of Sale, High Yield, Disease Resistance, and Days to Maturity. The Older, Large Household in Business (Cohort C) had significant results for Ease of Sale, High Yield, Least BW Damage, Disease Resistance, Days to Maturity, and Taste When Cooked. In contrast, the Older, Large Household Not in Business (Cohort D) showed significance only for Ease of Sale.

Across all cohorts, which included 189 farmers, High Yield, Disease Resistance, Ease of Sale, Resistance to BW, Days to Maturity, and Taste When Cooked emerged as key factors. Non-significant traits were Drought Tolerance, Pest Resistance, Dormancy, and Germination. Model fit improved significantly across cohorts and in the pooled sample (χ^2, $p < 0.05$), supporting result robustness.

3.6. Trait selection dynamics among commercial and subsistence potato farmers

Principal Component Analysis (PCA) delineated distinct trait preference structures across farmer cohorts (Fig. 8). The first two principal

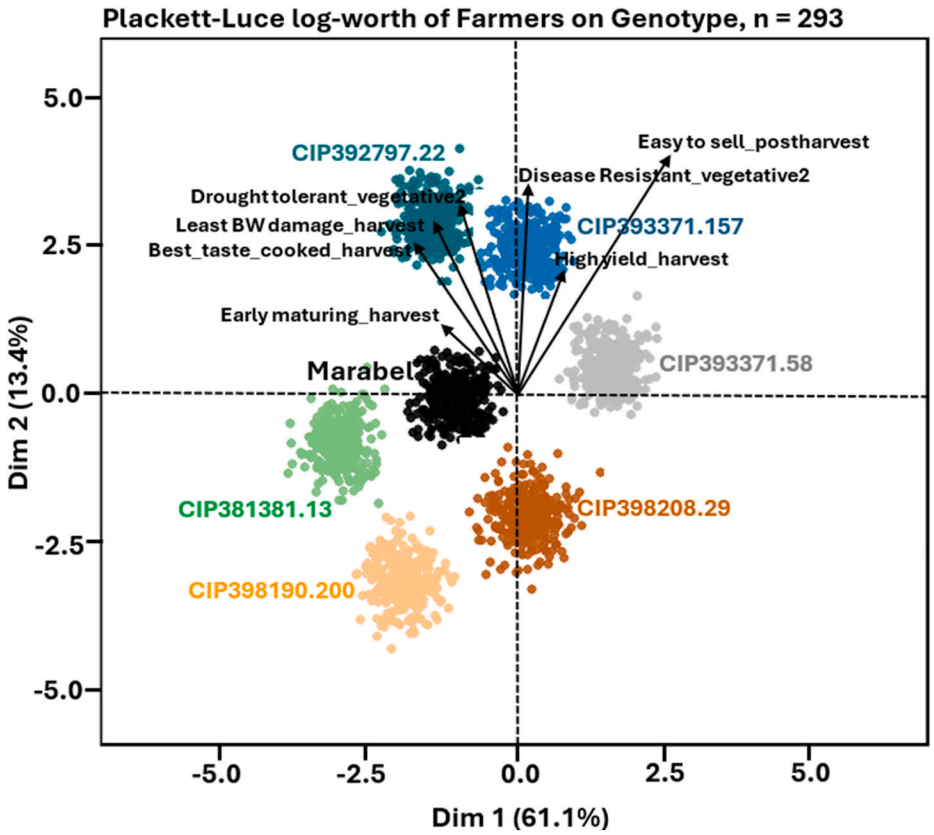


Fig. 6. Principal components of Plackett-Luce model estimates (log-worth) for potato genotype performance. Clusters indicate farmers' rankings, while vectors show correlations of genotypes retained after backward selection.

components collectively accounted for between 47 % and 58 % of the total variance within each cohort—34.1 % and 16.8 % for Cohort A (younger, in business), 27.9 % and 13.3 % for Cohort B (younger, not in business), 36.4 % and 16.7 % for Cohort C (older, in business), and 38.9 % and 19.3 % for Cohort D (older, not in business), capturing the major dimensions of variation in trait selection.

Robustness of these components was established through permutation tests (n = 1000), which confirmed that the observed eigenvalues for PC1 and PC2 significantly exceeded those expected under the null hypothesis ($p < 0.05$). Bootstrap resampling further substantiated the stability of trait loadings, with key traits demonstrating >90 % consistency across replicates, validating the reliability of trait associations in the PCA space.

Subgroup analyses revealed cohort-specific patterns: Cohort A, consisting of younger farmers who were engaged in the potato business (n = 24), exhibited a focused preference primarily for CIP392797.22. Their choices aligned strongly with market traits, emphasizing ease of sale at postharvest, high yield at harvest, and least BW damage at harvest.

In contrast, Cohort B, which included younger farmers not involved in the potato business (n = 17), demonstrated a broader selection that integrated both agronomic and market traits from a wider list of genotypes: CIP392797.22, CIP393371.157, and CIP393371.58. This group primarily focused on high yield at harvest while also recognizing the significance of agronomic traits such as drought tolerance and disease resistance. Their preference for Marabel stemmed from its early maturity, while CIP393371.58 was favored for its ease of sale postharvest. These two genotypes were positioned in a neighboring quadrant to CIP392797.22 and CIP393371.157.

Cohort C, comprising older farmers in the potato business (n = 108)-the dominant group, displayed a more subtle discriminatory approach by selecting four genotypes (CIP392797.22, CIP393371.157, CIP393371.58 and Marabel) with a greater emphasis on CIP392797.22 and CIP393371.157. Their preferences were skewed toward market traits, with the strongest alignments to ease of sale at postharvest, followed by best taste when cooked, high yield, and least BW damage.

Conversely, Cohort D, consisting of older farmers not engaged in the potato business (n = 40), exhibited flexibility in their selections across five genotypes, emphasizing agronomic traits such as disease resistance and drought tolerance at vegetative stage 2. Their mild preference for CIP381381.13 was linked to its ease of sale at postharvest stage.

Notably, CIP398190.200 and CIP398208.29 received no preference across cohorts, with trait vectors diverging away from these genotypes.

3.7. Genotypic trait selection in the potato business: A commercial farming perspective

Business involvement emerged as a stronger determinant of trait preference than age or household size. Accordingly, the younger and older age cohorts were aggregated, resulting in two analytically distinct groups among the 189 farmers: the Potato Business Cohort (n = 132) and the Non-Potato Business Cohort (n = 57). (Fig. 9). This subgroup analysis revealed contrasting patterns of trait selection between commercially engaged and non-engaged farmers. The Non-Potato Business Class, made their selection from a comprehensive array of traits drawn across vegetative, harvest, and post-harvest stages-demonstrating their flexibility. The traits consistently retained across iterations for this cohort included ease of sale at post-harvest, yield, and BW damage at harvest, as well as disease resistance, taste, drought resistance, early maturity, and germination at the vegetative stage. For genotype CIP392797.22, this cohort of farmers exhibited a wide range of traits across different crop growth stages. These included agronomic traits (drought tolerance, disease resistance at the vegetative stage, and early maturity at harvest) and market viability traits (high yield,

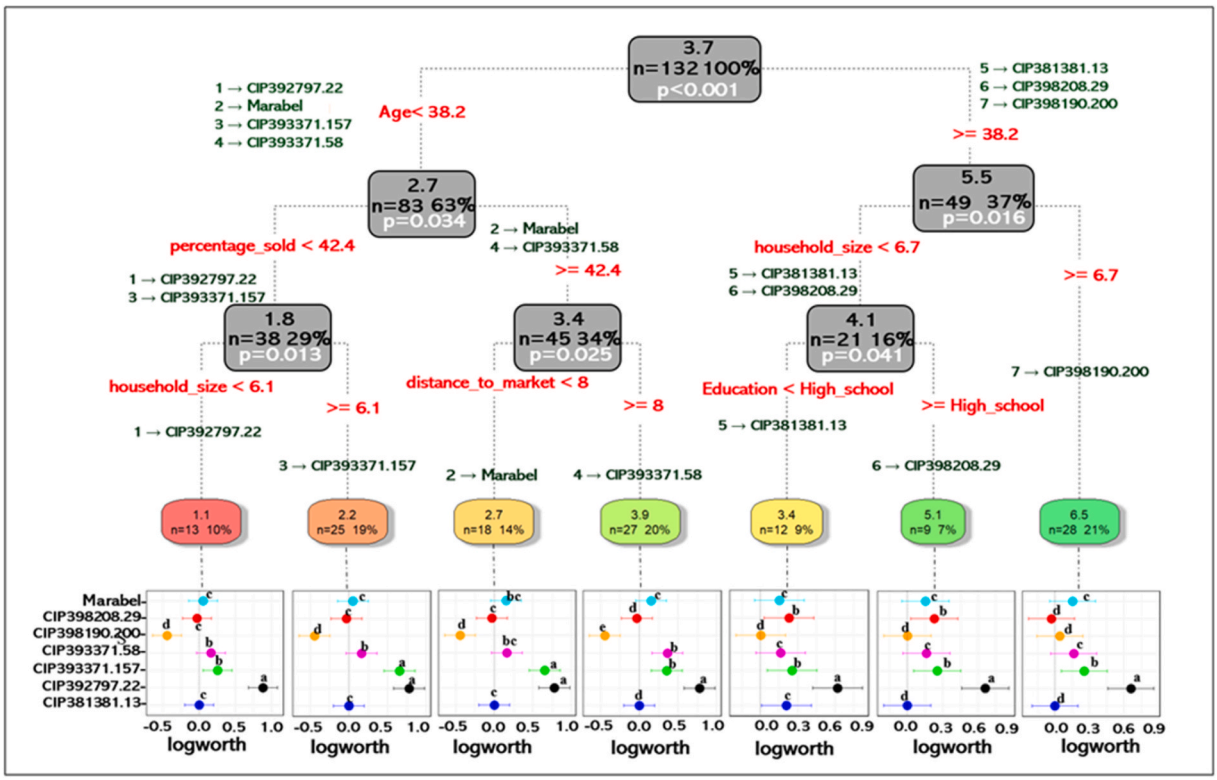


Fig. 7. CART Regression Analysis of Genotype Preferences. *Significant model parameters were identified with a threshold of $p < 0.05$, indicating the genotypes most influenced by each terminal node. Each terminal node corresponds to a specific genotype, with log-worth values indicating its preference level, while quasi-error bars illustrate the variability of influence within each node. These terminal nodes are the result of interconnections along the decision pathway, allowing each cut-off feature to be traced back, thus enabling a clear deduction of the principal factors influencing preference patterns for each genotype.*

Table 2
Segmentation of farmer cohorts by age, household size, and business involvement.

Cohort	Age Group	Household Size	Business Involvement	Total Count	Percentage of Total (%)
A. Younger, Small Household in Business	≤38.2 years	Small (≤6.7 members)	In Business (≥42.4 % sold)	24	8
B. Young, Small Household Not in Business	≤38.2 years	Small (≤6.7 members)	Not in Business (≤42.4 % sold)	17	6
C. Older, Large Household in Business	>38.2 years	Large (>6.7 members)	In Business (≥42.4 % sold)	108	37
D. Older, Large Household Not in Business	>38.2 years	Large (>6.7 members)	Not in Business (≤42.4 % sold)	40	14
Total				189	64

This segmentation is based on the cutoffs established by the CART regression analysis, identifying critical factors-household size (6.7 members), age (38.2 years), percentage of potatoes sold (42.4 %), and distance to market (8 km), all which significantly explain variations in farmer preferences. Both distance to market and percentage sold were considered factors of business involvement due to their strong correlation. Education did not show a significant impact when considered as a segment and, therefore, was not included in the aggregation.

minimal BW damage at harvest, ease of sale, and good taste when cooked postharvest). No trait was diverted to the worst bracket for this genotype.

Genotypes CIP393371.157, CIP393371.58, and Marabel were primarily selected for their high yield but faced challenges with ease of sale at postharvest stage. Marabel suffered from low disease resistance. CIP398190.200 performed worst overall, primarily due to poor germination, despite good tuber size in the few established stands. Similarly, CIP398208.29 ranked poorly across all traits, reinforcing its low preference. In contrast, CIP381381.13 performed negatively in most traits but was noted for ease of sale at post-harvest. For the Potato Business cohort, the selection criteria were highly discriminatory, focusing primarily on market viability traits. The weighting criteria favored ease of sale at postharvest, which negatively impacted most genotypes, relegating them to the worst category for this trait, except for CIP392797.22 and CIP381381.13. Restrictive yield criteria placed three genotypes—CIP398208.29, CIP381381.13, and CIP398190.200, into the worst bracket. The only reference to the vegetative stage in this cohort was CIP398190.200's poor germination, which contributed to its lowest yield score. Statistical robustness was supported by bootstrap resampling, which showed >90 % stability in top-ranking traits across cohorts. Permutation tests from PCA (see Section 3.6) indicated that cohort-specific separation exceeded random expectation ($p < 0.05$), validating the distinct preference structures.

4. Discussion

While national breeding and variety release programs continue to push improved potato varieties into the farming landscape, fundamental questions remain: are farmers truly at the center of these initiatives? Who are these varieties being bred for, and for what purpose are they released? Do these programs genuinely reflect the diverse realities of farmers—their demographics, market segments, and socio-economic contexts? If farmers are not meaningfully involved in breeding and variety selection, can these processes truly claim to address their needs,

Table 3
Log-likelihood ratio test estimates for primary traits evaluated by farmers within cohorts and overall.

Cohort	Trait	Deviance	Pr (>Chisq)
A. Younger, Small Household in Business (n = 24)	Ease of Sale at Postharvest	65.239	0.004*
	High Yield at Harvest	72.128	0.002*
	Least BW Damage at Harvest	58.439	0.021*
B. Younger, Small Household Not in Business (n = 17)	Ease of Sale at Postharvest	56.412	0.025*
	High Yield at Harvest	60.894	0.015*
	Disease Resistance at Vegetative	62.123	0.008*
	Days to Maturity at Harvest	45.181	0.019*
C. Older, Large Household in Business (n = 108)	Ease of Sale at Postharvest	80.342	0.001*
	High Yield at Harvest	85.967	0.000*
	Least BW Damage at Harvest	79.345	0.003*
	Disease Resistance at Vegetative	70.223	0.007*
	Days to Maturity at Harvest	88.301	0.000*
	Taste When Cooked at Postharvest	90.234	0.000*
D. Older, Large Household Not in Business (n = 40)	Ease of Sale at Postharvest	66.503	0.010*
Overall (n = 189)	High Yield at Harvest	72.398	0.002*
	Disease Resistance at Vegetative	63.894	0.008*
	Ease of Sale at Postharvest	72.349	0.003*
	High Yield at Harvest	75.727	0.001*
	Least BW Damage at Harvest	70.478	0.004*
	Disease Resistance at Vegetative	70.743	0.003*
	Drought Tolerance at Vegetative	64.124	0.514 ns
	Pest Resistance at Vegetative	56.102	0.296 ns
	Dormancy Duration at Postharvest	50.116	0.529 ns
	Days to Maturity at Harvest	78.675	0.000*
	Taste When Cooked at Postharvest	85.789	0.000*
	Germination at Early Vegetative	43.213	0.497 ns

Significance levels: "*" ($p \leq 0.05$), "ns" ($p > 0.05$), based on chi-square tests.

and will they adopt varieties that fail to do so?

The results of this study provide critical answers to these questions, demonstrating the value of participatory approaches, exemplified here by the tricot method, which empowers farmers to rank varieties based on real-world performance [42]. By allowing farmers to assess varieties under their own growing conditions and priorities, this approach ensures breeding efforts align with their actual needs, rather than being dictated solely by external selection criteria [31].

Across all growth stages—vegetative, maturity, and postharvest, three genotypes consistently emerged as superior performers: CIP392797.22, CIP393371.157, and CIP393371.58. These lines combined early vigor, strong germination, and drought and disease resilience with high yields, market-preferred traits, and excellent postharvest storability. Their robust performance across diverse selection points reflected a well-balanced trait portfolio that aligned with farmers' evolving priorities throughout the crop cycle. Notably, these genotypes are part of the nationally released variety pipeline [5], reflecting a convergence between farmer preferences and formal breeding objectives. However, this alignment does not negate the value of participatory approaches, which capture critical nuances often overlooked by top-down evaluations. For instance, genotype CIP398190.200, despite its official release, was less preferred by farmers primarily due to dormancy-related poor germination—a trait crucial for early-stage adoption but rarely incorporated in institutional trials.

Despite being the check, Marabel ranked low highlighting how farmer choices also hinge on context and behavior, not just traits. While initially valued for its early maturity, which typically aligns with market timing, its short dormancy, poor storability, and limited postharvest utility reduced its overall appeal to farmers. These trade-offs suggest that farmer preferences are shaped not only by observable field traits but also by behavioral factors such as risk aversion and planning for storage, household consumption, or market timing. Recognizing these dynamics in participatory breeding can explain misalignments between farmer choices and officially endorsed varieties, revealing a persistent gap in breeding approaches that prioritize top-down metrics over localized decision heuristics.

These shifting selection dynamics were significantly shaped by socioeconomic factors, which influenced farmers' evolving genotype preferences, aligning with Stringer et al. [43], who emphasized that different types of farmers require distinct adaptation and development options. Household size, age, market involvement, and proximity to markets played a crucial role in determining not only the prioritization of traits such as yield, disease resistance, and storability but also the balance between commercial production and subsistence farming.

Among market-oriented farmers, including younger individuals with small households and older farmers with larger households but engaged in business, high-yielding genotypes with strong postharvest and market traits were prioritized. CIP392797.22 and CIP393371.157 stood out in this cohort, combining productivity, bacterial wilt resistance, and storability—traits that enhanced market flexibility and income potential. Younger farmers favored varieties with strong market acceptance, while older ones emphasized consumer-preferred traits like cooking quality, recognizing their role in sustaining repeat sales. Similar trends are reported in Kenya and Ethiopia, where commercially engaged farmers prioritize marketability and disease resistance to maximize profitability [18,44].

In contrast, farmers not involved in potato sales—both younger and older with large households, prioritized food security, favoring genotypes that ensured a stable household supply. Their preferences leaned toward early maturity, drought tolerance, and disease resistance. Fast-maturing, resilient varieties were particularly attractive to younger food-security-focused farmers seeking reliable harvests under erratic rainfall. Older subsistence-oriented farmers prioritized consistency in yield and resistance to production shocks, with storability and postharvest traits considered secondary. These patterns are consistent with findings from Malawi, where non-commercial farmers prioritized food-secure traits over market traits [45].

As shown in the CART regression analysis of genotype preferences, education and business orientation played a complementary role in shaping farmers' choices. Those with higher education and business experience prioritized storability and market appeal, recognizing that the ability to store produce longer extends shelf life and allows farmers to delay sales until market conditions are favorable. This capacity helps them avoid immediate sales to brokers, who often pay lower prices, thus improving their bargaining power and potential income [46]. Their greater access to information and market awareness supports more strategic decision-making during variety evaluation. In contrast, farmers with limited education and no business focus emphasized early maturity, disease resistance, and yield to ensure household food security, relying more on traditional knowledge and immediate practical needs.

These findings underscore the critical role of socio-economic context in shaping farmer decisions, emphasizing the need to integrate these factors into variety development and dissemination strategies [47]. Unlike conventional approaches that categorize farmers based on singular factors such as gender, age, or production purpose [30,48,49], a

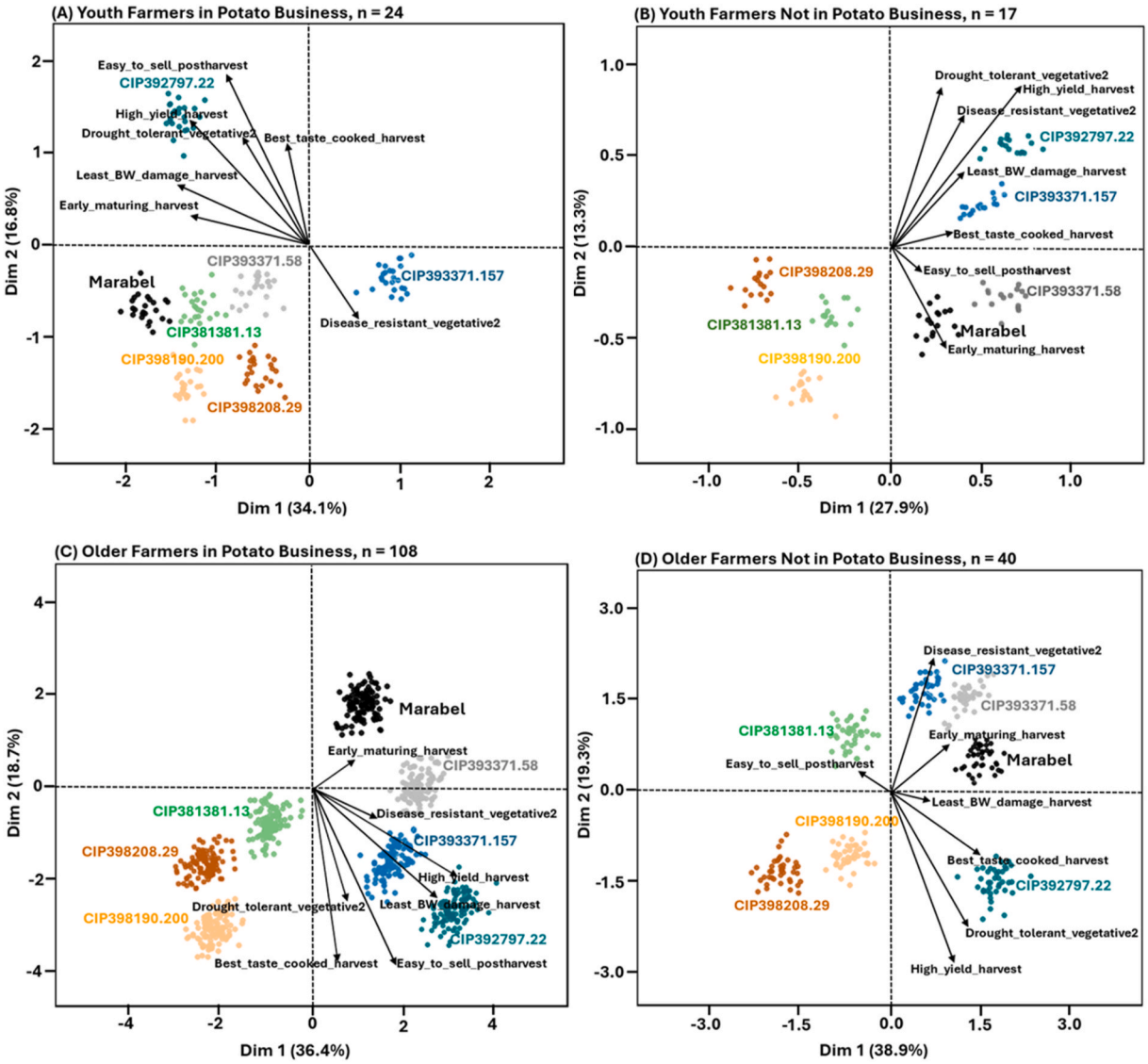


Fig. 8. Biplots of Plackett-Luce Model Estimates (log-worth) on Farmers' Cohorts (A. Younger, Small Household in Business; B. Younger, Small Household Not in Business; C. Older, Large Household in Business; D. Older, Large Household Not in Business). Clusters indicate farmers' rankings, while vectors show correlations of genotypes retained after backward selection.

more nuanced socio-economic segmentation enables breeding programs to tailor genotypes to specific demographic groups, ultimately improving adoption and impact [36]. Similar insights from other studies highlight the dynamic nature of farmer preferences, reinforcing the importance of aligning formal evaluations with real-world conditions to prevent the release of poorly adapted varieties and address the limitations of top-down breeding systems [29,32].

Ultimately, a participatory approach of this nature calls for a paradigm shift in how breeding and seed systems are structured, advocating for the prioritization of farmer input over researcher-breeder decisions. Such a shift necessitates a re-evaluation of agricultural policies to ensure they serve the diverse needs of farmers who are central to national food security. By recognizing farmers as key stakeholders in the breeding process, national policies can foster a more inclusive, equitable, and sustainable agricultural system. In this way, participatory research is not merely about improving varietal selection but transforming breeding systems to better serve the communities that rely on them, ultimately fostering a more resilient and food-secure future.

5. Conclusion

This study demonstrates the effectiveness of the tricot (triadic comparisons of technologies) approach in capturing complex, context-specific farmer preferences under farmer-managed conditions. It enabled scalable, high-resolution feedback across diverse environments, revealing how varietal priorities shift across growth stages and socio-economic profiles. Traits including drought tolerance, disease resistance, yield, storability, and marketability emerged distinctly, with genotypes CIP392797.22, CIP393371.157, and CIP393371.58 consistently favored, while others were rejected due to poor agronomic or post-harvest performance. These insights underscore the capacity of farmer-led evaluations to generate granular, decision-relevant data that challenge assumptions often embedded in conventional breeding.

By situating varietal assessment within the lived realities of smallholder systems and layering it with digital and analytical tools, this approach offers a compelling model for rethinking how varieties are developed and deployed. It points toward more responsive, demand-driven breeding pathways that better reflect the heterogeneity of farmer needs and production conditions. Yet, scaling such frameworks

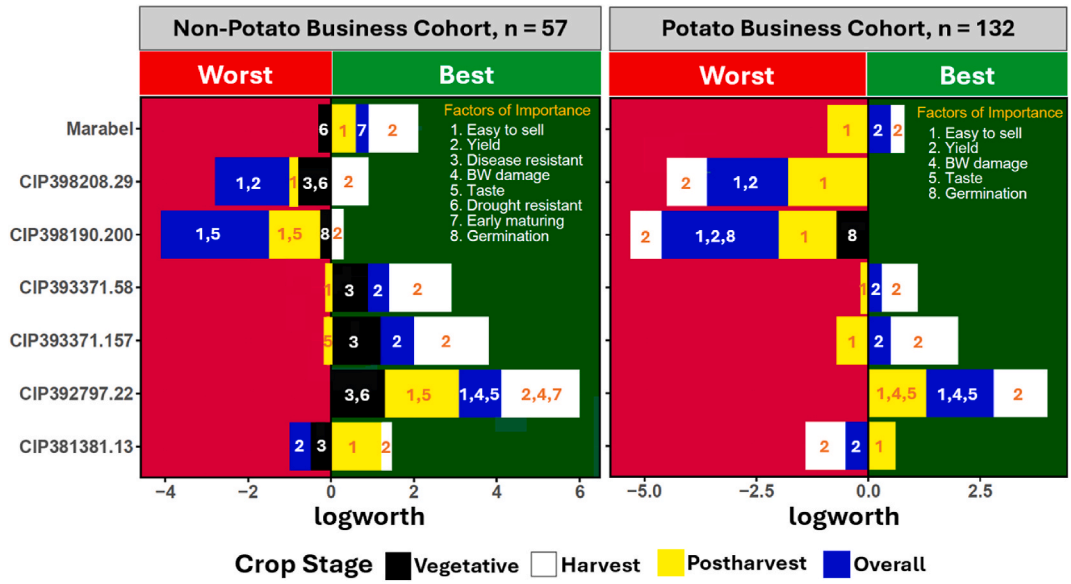


Fig. 9. Potato Business Engagement and Genotype Preferences Across Evaluation Stages on a Best-Worst Log-worth Scale. The codes correspond to traits listed under the "factor of importance."

will require overcoming constraints related to digital access, institutional inertia, and system integration. Even so, the study illustrates how aligning technical innovation with participatory practice can unlock new possibilities for varietal targeting, ultimately contributing to more resilient and equitable agricultural systems.

CRediT authorship contribution statement

K. Sharma: Writing – original draft, Resources, Investigation, Formal analysis, Conceptualization, Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation. **E. Atieno:** Writing – review & editing, Methodology, Conceptualization, Writing – original draft, Investigation. **J. Mugo:** Writing – original draft, Writing – review & editing, Data curation. **K. de Sousa:** Software, Conceptualization, Writing – review & editing, Methodology. **J. van Etten:** Software, Conceptualization, Writing – review & editing, Formal analysis. **S. Nyawade:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization, Writing – original draft, Validation, Investigation, Data curation.

Research ethics statement

This study adhered to ethical guidelines for research involving human participants. Oral informed consent was obtained and documented at the start of the survey, with enumeration stopping if permission was denied. Participants were informed of the study's purpose, procedures, risks, and their rights to confidentiality and voluntary participation, including the option to skip questions. Data were anonymized, and no vulnerable populations were involved. Ethical approval was granted by the project leader at the International Potato Center (CIP).

Declaration of competing interest

The authors have no competing interests to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jafr.2025.102135.

Data availability

Data will be made available on request.

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